ROUSSEL Victor

Paris Area, France
victor.a.roussel@gmail.com | +33 7 60 11 70 19 | https://vicosr.github.io/ | victor-a-roussel | Vicosr

Second-year engineering student at CentraleSupélec with a strong interest in applied mathematics, particularly in AI, data science, finance, cryptography, and computer science. Actively seeking internships in these fields.

EDUCATION

CentraleSupélec - Université Paris-Saclay — **Paris-Saclay, France**
Engineering Degree — Sep 2024 – Jun 2027
- GPA: 3.92/4.33 (Year 1). Key courses: Measure Theory & Probability (Hilbert spaces, L^p spaces, conditional expectation, random walks), Partial Differential Equations (distributions, Sobolev spaces, variational formulation), Algebra & Cryptology (lattice reduction, LLL algorithm, elliptic curves, learning with errors), Information Theory (entropy, mutual information, f-divergences), Statistics & Machine Learning, Optimization (convexity, duality, Lagrange multipliers)

Lycée Hoche — **Versailles, France**
CPGE: MPSI – MP — Sep 2021 – Jun 2024
- Intensive two-year undergraduate program in mathematics and physics, preparing for nationwide competitive entrance exams to top French engineering schools (equivalent to a Bachelor's level in Mathematics).
- Subjects: Mathematics, Physics, Chemistry, Computer Science

PROFESSIONAL EXPERIENCE

Lycée Hoche - Éducation Nationale — **Versailles, France**
Oral Examiner — Sep 2024 – Jun 2026
- Assessed 6 second-year students weekly in 2-hour oral examinations ("khôlles") in mathematics, preparing tailored exercises for each student's level (see examples) and reporting individual progress to the professor.

Sumécatronic — **Magny-Les-Hameaux, France**
Milling Machine Operator — Jun 2025 – Jul 2025
- Autonomous milling machine operator, producing and quality-controlling series of precision-engineered parts for the aeronautics industry. Gained hands-on experience with quality assurance processes and workshop logistics.

PROJECTS

Course project: Variance Reduction for Monte Carlo Simulation in Financial Mathematics | — **Gif-sur-Yvette, France**
Contributor — Mar 2026
- Explored low-discrepancy sequences as alternatives to standard Monte Carlo methods and built a benchmark comparing all approaches on a grid of volatility, dimension, and moneyness using $|\text{err}|\sqrt{\text{time}}$ as a metric.
- Found that Sobol sequences were the most efficient method in 46% of configurations; trained a random forest achieving 81% CV accuracy to predict the best method for a given configuration.

Course project: Multi-view Sport Video Synchronisation — GoPro | — **Gif-sur-Yvette, France**
Contributor — Jan 2026 – Present
- Exploring methods for automatic multi-view sport video synchronisation, comparing three approaches: human pose detection, audio-based alignment using transformers, and visual similarity matching.

Course project: Robot Fault Diagnosis through Transfer Learning and Digital Twins — CentraleSupélec LGI | Gif-sur-Yvette, France
Contributor — Jan 2025 – Jun 2025
- Since failure data is scarce for mechanical arms (built not to fail), explored the feasibility of classifying real faults using a classifier trained on simulated data — a typical domain adaptation problem.
- Achieved 87% AUC on real robot fault detection using a DANN (Domain-Adversarial Neural Network) built on a 1D-CNN architecture.

LEADERSHIP & INVOLVEMENT

Handi-tutorat (in partnership with Sopra Steria) — **Gif-sur-Yvette, France**
Tutoring for students with disabilities — Jan 2025 – Present
- Volunteered to support students with disabilities facing academic difficulties, helping them develop personalized learning methods and integrate into campus life.

Private Lessons — **Paris Area, France**
Mathematics Tutor — Sep 2024 – Jun 2025
- Provided academic support for high schoolers in mathematics, focusing on building their mathematical intuition and fostering independence.

Cooking Association of CentraleSupélec — **Gif-sur-Yvette, France**
Head of Testing — Sep 2024 – Present
- Organise and lead weekly tasting sessions to evaluate and select recipes for association events.

SKILLS

Programming: Python (NumPy, Pandas, scikit-learn, PyTorch), OCaml, SQL, MATLAB, R, Excel;

Tools: Git, Bash, Slurm, HPC;

Languages: French (native), English (advanced), German (intermediate).

INTERESTS

Golf: Regular player;

Harp: 10 years, 7 years in conservatoire;

Associations: Cooking association of CentraleSupélec;